Stochastic Processes

Homework 2: Complete Step-by-Step Solutions

Lecture 2 | 20 problems | 100 points

Reading alignment

Sheldon M. Ross, *Introduction to Probability Models*, 10th ed.

- **Primary:** Chapter 2: random variables, distributions, moments, covariance, and joint distributions.
- **Supporting:** Law of large numbers and central limit theorem from lecture notes.

Instructor note. Solutions correspond exactly to the distributed Homework 2, Problems 1–20, in their original order. Intermediate steps are shown explicitly. Decimal probabilities are rounded to four places when appropriate. Problems 19 and 20 extend the basic moment identities to random sums and diversification.

I. Random Variables and Distributions — 16 points

Problem 1. A discrete probability law (4 points)

(a) Normalization gives $1 = \sum_{k=0}^{4} c(k+1) = c(1+2+3+4+5) = 15c$, so $\boxed{c = 1/15}$.

(b) $P(X \geq 3) = p_X(3) + p_X(4) = 4/15 + 5/15 = \boxed{3/5 = 0.6}$.

(c) Weight each support value by its probability:

$$E[X] = \frac{1}{15}\sum_{k=0}^{4} k(k+1) = \frac{0+2+6+12+20}{15} = \boxed{\frac{8}{3}}.$$

(d) The second moment is

$$E[X^2] = \frac{1}{15}\sum_{k=0}^{4} k^2(k+1) = \frac{0+2+12+36+80}{15} = \boxed{\frac{26}{3}}.$$

Therefore $\mathrm{Var}(X) = 26/3 - (8/3)^2 = (78-64)/9 = \boxed{14/9}$.

Problem 2. A cumulative distribution function (4 points)

(a) Accumulate the masses at $-2, 0, 1, 3$:

$$\boxed{F_X(x) = \begin{cases} 0, & x < -2, \\ 0.1, & -2 \leq x < 0, \\ 0.3, & 0 \leq x < 1, \\ 0.7, & 1 \leq x < 3, \\ 1, & x \geq 3. \end{cases}}$$

(b) The values satisfying $-1 < X \le 1$ are 0 and 1, so $\boxed{P(-1 < X \le 1) = 0.2 + 0.4 = 0.6}$.

(c) $E[X] = (-2)(0.1) + 0(0.2) + 1(0.4) + 3(0.3) = \boxed{1.1}$.

(d) **No.** The expectation is a weighted average, not necessarily a realizable outcome. Here 1.1 is not in the support $\{-2, 0, 1, 3\}$.

Problem 3. A continuous density (4 points)

(a) Require unit area: $1 = c\int_0^2 x^2\,dx = c[x^3/3]_0^2 = 8c/3$, hence $\boxed{c = 3/8}$.

(b) For $0 \le x \le 2$, integrate the density:

$$F_X(x) = \int_0^x \frac{3}{8}t^2\,dt = \frac{x^3}{8}, \qquad \boxed{F_X(x) = \begin{cases} 0, & x < 0, \\ x^3/8, & 0 \le x \le 2, \\ 1, & x > 2. \end{cases}}$$

(c) By the complement rule, $P(X > 1) = 1 - F_X(1) = 1 - 1/8 = \boxed{7/8}$.

(d) $E[X] = \int_0^2 x(3x^2/8)\,dx = (3/8)[x^4/4]_0^2 = \boxed{3/2}$.

Problem 4. Binomial and Poisson moments (4 points)

(a) Binomial formulas: $\boxed{E[X] = 20(0.3) = 6}$ and $\boxed{\mathrm{Var}(X) = 20(0.3)(0.7) = 4.2}$.

(b) Poisson formulas: $\boxed{E[Y] = 4}$ and $\boxed{\mathrm{Var}(Y) = 4}$.

(c) Since $E[Z^2] = \mathrm{Var}(Z) + E[Z]^2$,

$$\boxed{E[X^2] = 4.2 + 36 = 40.2}, \qquad \boxed{E[Y^2] = 4 + 16 = 20}.$$

(d) The difference is precisely variance: $E[X^2] - (E[X])^2 = \mathrm{Var}(X) = 4.2$. They agree only if X is constant almost surely.

II. Expectation and Variance

24 points

Problem 5. Moments from first principles (6 points)

(a) Sum values and squared values weighted by their masses:

$$E[X] = (-2)(0.1) + 0(0.3) + 1(0.4) + 4(0.2) = \boxed{1},$$
$$E[X^2] = 4(0.1) + 0 + 1(0.4) + 16(0.2) = \boxed{4}.$$

(b) $\mathrm{Var}(X) = E[X^2] - E[X]^2 = 4 - 1^2 = \boxed{3}$.

(c) The standard deviation is $\boxed{\sigma_X = \sqrt{3} \approx 1.7321}$.

(d) Directly,

$$E[(X-1)^2] = 9(0.1) + 1(0.3) + 0(0.4) + 9(0.2) = 0.9 + 0.3 + 1.8 = \boxed{3}.$$

This equals $\mathrm{Var}(X)$ because $1 = E[X]$: variance is the expected squared distance from the mean.

Problem 6. Affine transformations (6 points)

(a) Linearity yields $E[Y] = 3E[X] - 4 = 15 - 4 = \boxed{11}$ and $E[Z] = -2E[X] + 7 = -10 + 7 = \boxed{-3}$.

(b) Constants do not contribute to variance, and scale factors are squared:

$$\boxed{\mathrm{Var}(Y) = 3^2(9) = 81}, \qquad \boxed{\mathrm{Var}(Z) = (-2)^2(9) = 36}.$$

(c) Rearrange the variance identity: $E[Y^2] = \mathrm{Var}(Y) + E[Y]^2 = 81 + 121 = \boxed{202}$.

(d) For any b, $(X + b) - E[X + b] = X - E[X]$. The deviations from the mean are unchanged, so $\mathrm{Var}(X + b) = \mathrm{Var}(X)$.

Problem 7. Deriving variance identities (6 points)

(a) Expand the square and use linearity:

$$\begin{aligned}\mathrm{Var}(X) &= E[(X-\mu)^2] \\ &= E[X^2 - 2\mu X + \mu^2] \\ &= E[X^2] - 2\mu E[X] + \mu^2 \\ &= E[X^2] - 2\mu^2 + \mu^2 = \boxed{E[X^2] - \mu^2}.\end{aligned}$$

(b) $E[aX + b] = a\mu + b$, hence

$$\begin{aligned}\mathrm{Var}(aX + b) &= E[((aX + b) - (a\mu + b))^2] \\ &= E[a^2(X-\mu)^2] = \boxed{a^2 \,\mathrm{Var}(X)}.\end{aligned}$$

(c) Set $a = -1, b = 0$: $\mathrm{Var}(-X) = (-1)^2 \,\mathrm{Var}(X) = \boxed{\mathrm{Var}(X)}$. Reflection changes the signs of deviations, not their squares.

Problem 8. Linearity without independence (6 points)

(a) $E[N] = E[X_1] + E[X_2] + E[X_3] = 8 + 12 + 10 = \boxed{30}$ arrivals.

(b) $E[R] = 15E[N] = 15(30) = \boxed{\$450}$.

(c) $E[\Pi] = E[R] - 120 = 450 - 120 = \boxed{\$330}$.

(d) **No.** Linearity of expectation holds for dependent random variables as long as their expectations exist. Independence would matter when simplifying the variance of the sum.

III. Covariance and Joint Distributions 30 points

Problem 9. Covariance from a joint PMF (6 points)

(a) Sum across rows for X and columns for Y:

$$p_X(0) = 0.2 + 0.3 = \boxed{0.5}, \quad p_X(1) = 0.1 + 0.4 = \boxed{0.5},$$

$$p_Y(0) = 0.2 + 0.1 = \boxed{0.3}, \quad p_Y(1) = 0.3 + 0.4 = \boxed{0.7}.$$

(b) Since the variables are binary, $E[X] = P(X = 1) = \boxed{0.5}$ and $E[Y] = P(Y = 1) = \boxed{0.7}$. The product XY is 1 only at $(1, 1)$, so $\boxed{E[XY] = 0.4}$.

(c) $\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0.4 - (0.5)(0.7) = \boxed{0.05}$.

(d) If independent, $P(X = 1, Y = 1)$ would equal $0.5(0.7) = 0.35$. The joint PMF gives 0.4, so **they are dependent**.

Problem 10. Variance of sums and differences (6 points)

(a) $E[2X - Y] = 2(4) - 7 = \boxed{1}$.

(b) $\mathrm{Var}(X + Y) = 9 + 16 + 2(-3) = \boxed{19}$.

(c) For constants a, b, $\mathrm{Var}(aX + bY) = a^2 \mathrm{Var}(X) + b^2 \mathrm{Var}(Y) + 2ab \mathrm{Cov}(X, Y)$. Thus

$$\mathrm{Var}(2X - Y) = 4(9) + 16 + 2(2)(-1)(-3) = \boxed{64}.$$

(d) Under independence covariance would be zero and the sum variance would be $9 + 16 = 25$. The negative covariance lowers it to 19.

Problem 11. Correlation and its interpretation (6 points)

(a) $\sigma_X = 5$, $\sigma_Y = 6$, so $\rho_{X,Y} = 18/(5 \cdot 6) = \boxed{0.6}$.

(b) Translation does not affect covariance; scaling multiplies it by both coefficients:

$$\mathrm{Cov}(3X + 2, -2Y + 5) = 3(-2)(18) = \boxed{-108}.$$

(c) Positive scaling of X and negative scaling of Y reverse the sign of correlation, leaving its magnitude unchanged: $\boxed{\mathrm{Corr}(3X + 2, -2Y + 5) = -0.6}$.

(d) Zero covariance means no *linear* co-movement. Independence implies zero covariance when moments exist, but zero covariance does not imply independence; nonlinear dependence can remain.

Problem 12. A dependent random-variable construction (6 points)

(a) With each value of X equally likely, $E[X] = (-1 + 0 + 1)/3 = \boxed{0}$. Since $Y = X^2$, $E[Y] = (1 + 0 + 1)/3 = \boxed{2/3}$. Also $XY = X^3$, so $E[XY] = (-1 + 0 + 1)/3 = \boxed{0}$.

(b) $\mathrm{Cov}(X, Y) = 0 - (0)(2/3) = \boxed{0}$.

(c) **Not independent.** For example, $P(Y = 0 \mid X = 0) = 1$ but $P(Y = 0) = 1/3$. Knowing X determines Y.

(d) The variables are dependent even though their covariance vanishes. Covariance measures linear association, not every form of dependence.

Problem 13. Continuous joint density (6 points)

(a) Normalize over the unit square:

$$1 = c \int_0^1 \int_0^1 (x + y)\, dy\, dx = c\left(\frac{1}{2} + \frac{1}{2}\right) = c, \quad \boxed{c = 1}.$$

(b) For $0 \le x, y \le 1$,

$$\boxed{f_X(x) = \int_0^1 (x + y)\, dy = x + \frac{1}{2}}, \quad \boxed{f_Y(y) = \int_0^1 (x + y)\, dx = y + \frac{1}{2}}.$$

Both marginal densities are zero outside $[0, 1]$.

(c) By symmetry,

$$E[X] = \int_0^1 x\left(x + \frac{1}{2}\right) dx = \frac{1}{3} + \frac{1}{4} = \boxed{\frac{7}{12}}, \quad \boxed{E[Y] = \frac{7}{12}}.$$

For the cross moment,

$$\begin{aligned} E[XY] &= \int_0^1 \int_0^1 xy(x + y)\, dy\, dx \\ &= \left(\int_0^1 x^2 dx\right)\left(\int_0^1 y\, dy\right) + \left(\int_0^1 x\, dx\right)\left(\int_0^1 y^2 dy\right) \\ &= \frac{1}{6} + \frac{1}{6} = \boxed{\frac{1}{3}}. \end{aligned}$$

(d) $\mathrm{Cov}(X, Y) = 1/3 - (7/12)^2 = (48 - 49)/144 = \boxed{-1/144}$. Because independence would force covariance zero (finite moments hold), the variables are **dependent**.

IV. Asymptotics and Limit Theorems 16 points

Problem 14. Mean and variance of a sample average (4 points)

(a) Linearity gives $E[\overline{X}_n] = n^{-1} \sum_{i=1}^n E[X_i] = n^{-1}(n\mu) = \boxed{\mu}$.

(b) In general, variance of a sum includes covariances. Independence makes all off-diagonal covariances zero:

$$\mathrm{Var}(\overline{X}_n) = \frac{1}{n^2} \sum_{i=1}^n \mathrm{Var}(X_i) = \frac{n\sigma^2}{n^2} = \boxed{\frac{\sigma^2}{n}}.$$

(c) $\mathrm{SD}(\overline{X}_{144}) = \sigma/\sqrt{144} = 6/12 = \boxed{0.5}$.

Problem 15. Law of large numbers and Chebyshev's inequality (4 points)

(a) Since $\mathrm{Var}(\overline{X}_n) = 25/n$, Chebyshev's inequality gives

$$P(|\overline{X}_n - 10| \geq 1) \leq \frac{25/n}{1^2} = \boxed{\frac{25}{n}}.$$

(The bound may exceed 1 for small n, in which case the trivial bound 1 is tighter.)

(b) Require $25/n \leq 0.05$, or $n \geq 25/0.05 = \boxed{500}$. Thus 500 observations suffice for this bound.

(c) The weak law states $\overline{X}_n \xrightarrow{P} 10$: for every $\varepsilon > 0$, $P(|\overline{X}_n - 10| > \varepsilon) \to 0$. Here the Chebyshev bound tends to zero as n grows, demonstrating the convergence.

Problem 16. Central limit theorem for averages (4 points)

(a) $E[\overline{X}_{100}] = \boxed{12}$ minutes and $\mathrm{Var}(\overline{X}_{100}) = 16/100 = \boxed{0.16}$ minutes squared. Its standard deviation is 0.4 minutes.

(b) By the CLT, $\overline{X}_{100}$ is approximately $N(12, 0.16)$. Standardize:

$$P(11.2 \leq \overline{X}_{100} \leq 12.8) \approx P\left(\frac{11.2 - 12}{0.4} \leq Z \leq \frac{12.8 - 12}{0.4}\right)$$

$$= \Phi(2) - \Phi(-2) = 2\Phi(2) - 1 = \boxed{0.9545}.$$

(c) The underlying service-time distribution is unspecified. The CLT gives a limiting normal approximation for large sample sizes, not an exact normal law at $n = 100$ (unless additional distributional assumptions hold).

Problem 17. Asymptotics for a sum (4 points)

(a) $E[S_{400}] = 400(3) = \boxed{1200}$ and $\mathrm{Var}(S_{400}) = 400(4) = \boxed{1600}$; the standard deviation is 40.

(b) By the CLT,

$$P(S_{400} > 1240) \approx P\left(Z > \frac{1240 - 1200}{40}\right) = 1 - \Phi(1) = \boxed{0.1587}.$$

(c) For i.i.d. observations with variance σ^2, $\mathrm{SD}(S_n) = \sigma\sqrt{n}$ grows like $\sqrt{n}$, while $\mathrm{SD}(\overline{X}_n) = \sigma/\sqrt{n}$ shrinks like $1/\sqrt{n}$. Their centers scale as $n\mu$ and μ, respectively.

V. Synthesis and Mathematical Reasoning 14 points

Problem 18. A common shock and correlated counts (4 points)

(a) By linearity and independence,

$$\boxed{E[X] = 2 + 4 = 6}, \quad \boxed{E[Y] = 3 + 4 = 7},$$

$$\boxed{\mathrm{Var}(X) = 1 + 3 = 4}, \quad \boxed{\mathrm{Var}(Y) = 2 + 3 = 5}.$$

(b) Expand covariance bilinearly:

$$\begin{aligned}\mathrm{Cov}(X,Y) &= \mathrm{Cov}(U+W, V+W)\\ &= \mathrm{Cov}(U,V) + \mathrm{Cov}(U,W) + \mathrm{Cov}(W,V) + \mathrm{Var}(W)\\ &= 0+0+0+3 = \boxed{3}.\end{aligned}$$

(c) $X - Y = (U+W) - (V+W) = U - V$, so independence gives $\mathrm{Var}(X-Y) = 1+2 = \boxed{3}$. The common shock W cancels exactly when the two counts are subtracted.

Problem 19. A random sum: conditional moments (5 points)

(a) Conditional on $N = n$, S sums n independent marks with mean 2 and variance 3:

$$E[S \mid N=n] = 2n, \qquad \mathrm{Var}(S \mid N=n) = 3n.$$

Therefore $\boxed{E[S \mid N] = 2N}$ and $\boxed{\mathrm{Var}(S \mid N) = 3N}$.

(b) Total expectation gives $E[S] = E[E[S \mid N]] = E[2N] = 2E[N] = 2(5) = \boxed{10}$.

(c) Total variance separates two terms:

$$\begin{aligned}\mathrm{Var}(S) &= E[\mathrm{Var}(S \mid N)] + \mathrm{Var}(E[S \mid N])\\ &= E[3N] + \mathrm{Var}(2N)\\ &= 3(5) + 4\,\mathrm{Var}(N)\\ &= 15 + 4(5) = \boxed{35}.\end{aligned}$$

(d) The first term (15) is variation in the sizes of the Z_i contributions for a given count. The second (20) is variation caused by the random number N of contributions.

Problem 20. Portfolio variance and diversification (5 points)

(a) By linearity,

$$\boxed{E[R_p] = 0.06w + 0.10(1-w) = 0.10 - 0.04w}.$$

Using the variance-of-a-linear-combination formula,

$$\begin{aligned}\mathrm{Var}(R_p) &= w^2(0.04) + (1-w)^2(0.09) + 2w(1-w)(0.03)\\ &= 0.04w^2 + 0.09(1-2w+w^2) + 0.06w - 0.06w^2\\ &= \boxed{0.07w^2 - 0.12w + 0.09}.\end{aligned}$$

(b) At $w = 1/2$, $E[R_p] = 0.10 - 0.04(0.5) = \boxed{0.08}$ and

$$\mathrm{Var}(R_p) = 0.07(0.25) - 0.12(0.5) + 0.09 = \boxed{0.0475}.$$

(c) Differentiate $V(w) = 0.07w^2 - 0.12w + 0.09$:

$$V'(w) = 0.14w - 0.12 = 0 \quad \Longrightarrow \quad \boxed{w^* = \frac{0.12}{0.14} = \frac{6}{7} \approx 0.8571}.$$

Since $V''(w) = 0.14 > 0$ and $w^* \in [0,1]$, this is the unique minimum on the permitted interval. The minimum variance is $V(6/7) = \boxed{27/700 \approx 0.0386}$.

(d) Covariance enters as $2w(1-w)\,\mathrm{Cov}(R_1, R_2)$. Lower or negative covariance reduces portfolio variance for nonnegative weights, increasing diversification benefits. Positive covariance reduces these benefits relative to uncorrelated assets.

End of solution guide. Every answer corresponds to the issued Homework 2 problem statements. Limit-theorem probabilities are approximations unless otherwise specified.